A Framework for Quality Enhancement of Multispectral Remote Sensing Images

Shilpa Suresh
Department of E&C Engineering,
National Institute of Technology
Karnataka, Surathkal, Mangaluru-
575025, Karnataka, India
shilparagesh89@gmail.com

Devikalyan Das
Department of E&T Engineering,
Veer SurendraSai University of
Technology, Burla, Sambalpur-
768018, Odisha, India
devikalyandas906@gmail.com

Shyam Lal
Department of E&C Engineering,
National Institute of Technology
Karnataka, Surathkal, Mangaluru-
575025, Karnataka, India
shyamfec@nitk.edu.in

***Abstract*—Researches in satellite image enhancement have been particularly confined to two major areas- contrast enhancement and image denoising of remote sensing images. The processing of relatively dark or shadowed images necessitates the need for robust remote sensing enhancement techniques. In this paper, a robust framework for quality enhancement of multispectral remote sensing images is proposed. The quantitative results of proposed algorithm and other existing remote sensing enhancement algorithms are calculated in terms of DE, NIQMC, BIQME, PisDist and CM on different remote sensing and other image databases. Results reveal that visual enhancement of the proposed algorithm is better than other existing remote sensing enhancement algorithms. Finally, the simulation experimental results show that proposed algorithm is effective and efficient for remotes sensing as well as natural images.**

Keywords—Remote sensing; Enhancement; Contrast.

I. INTRODUCTION

Satellite and aerial images are very popularly preferred in many disciplines such as defense and space department, geoscience studies, etc. But sometimes these images are degraded in terms of visual quality owing to many environmental and other internal and external miscellaneous factors like the noise during their acquisition. So the image processing and analysis application facilitates an improvement in the visual characteristics of such images. Contrast of an Image is one such characteristic that pitch in for the quality improvement of image perceived. Since human visual system is more sensitive towards contrast of image instead of its absolute luminance, it is considered as a disparity in the color and brightness of these objects. Low-contrast and high contrast regions appear to be dark and artificially illuminated respectively. Both of this will lead to the depletion of relevant information. So, optimal enhancement of the image contrast is done for depicting all the information present in the input image.

In this paper, we propose a modified contrast enhancement method based on a modified linking synaptic computation. Its primary focus is improving the contrast of the image using neural mechanisms and keeping in mind the preservation of image information. We have also compared the results with state-of-art remote sensing enhancement methods like Regularized Histogram Equalization method and Discrete Cosine Transform (RHE-DCT) [4], Transform domain method based on discrete wavelet transform and singular value decomposition (DWT-SVD) [5], Local color correction method (LCC) [6], Reinhard Tone mapping method (REINHARD) [7], Drago Tone mapping method (DRAGO) [8] and Linking Synaptics Computation Network (LSCN) [1].

II. PROPOSED FRAMEWORK

This section presents detailed analysis of proposed framework for quality enhancement of multispectral remote sensing Images. The proposed framework is explained in two phases. In the first phase modified linking synaptic computation network (MLSCN) and in the second phase modified contrast enhancement algorithm are explained.

A. MLSCN

The linking synapse, the membrane potential and the threshold are embodied as leaky integrator, which is explained in the [1]. The linking synapse is modulated by the postsynaptic potential feedback. The linking synapse is represented by a leaky integrator [16-19].

$$X_{ij}(n) = lX_{ij}(n-1) + \sum_{pq} W_{ijpq} Z_{pq}(n-1) \qquad (1)$$

Where each neuron is denoted with indices (i, j), one of its neighboring neurons is denoted with (p, q), l is the linking constant, W_{ijpq} is the weight applied to a linking synapse and $Z_{pq}(n-1)$ is the postsynaptic action potential.

Cortical networks consists of both feedback and feed-forward components. The membrane potential is produced by the combination of the aforementioned components and is depicted by a leaky integrator,

$$P_{ij}(n) = gP_{ij}(n-1) + S_{ij}[1 + \beta X_{ij}(n)] \qquad (2)$$

Where $g$is the membrane potential attenuation constant, Sij carries the stimulus information and β is the linking strength.

The threshold is an eduction from the neuron analog in [2]. With the help of a leaky integrator, the threshold of a neuron isrepresented as

$$\xi_{ij}(n) = f\xi_{ij}(n-1) + hZ_{ij}(n-1)] \quad (3)$$

Where f the threshold attenuation is constant, h is a magnitude adjustment, and $Z_{ij}(n-1)$ is the postsynaptic action potential.

$$z_{ij}(n) = \begin{cases} 1, & if\ P_{ij}(n) > \xi_{ij}(n) \\ 0, & otherwise \end{cases} \quad (4)$$

The modified linking synaptic computation network (MLSCN) is described by equation (1), (2), 3) and (4). MLSCN withholds all the significant properties as the LSCN. The alteration mainly focuses on the dynamic threshold.

When the membrane potential almost equals threshold then this gives the spike time.

$$\xi_{ij}(n) = P_{ij}(n) \quad (5)$$

After first spike is fired by a neuron, the threshold is modified in the successive iteration to

$$\xi_{ij}(n+1) = fP_{ij}(n) + h \quad (6)$$

After that there is exponential delay in the threshold as per the attenuation constant f. The firing of the second spike takes place when the membrane potential and the threshold again become almost equal to each other. Hence, a neuron's firing cycle is represented as,

$$C_{ij} = \log_f \left(\frac{P_{ij}}{fP_{ij} + h} \right) \quad (7)$$

The iterations for proposed model is stopped when all the neurons present in the network produce their corresponding spikes.

B. Modified Contrast Enhancement Algorithm

The proposed synaptic computation network is backed by a modified contrast enhancement techniques. But before applying the modified contrast enhancement technique, the image is smoothed and filtered by the guided filter [20].

a. Edge Enhancement Effect

The initial value of the threshold decides the start point of the linking synaptic integration. The low intensity pixels fire slower than the high intensity pixels as a result in successive terms, the higher intensity pixels may fire pixels which will be belonging to dark region. These may appear in the output image as noisy pixels. So the edge enhanced effect must be such that the firing of pixels occur in descending order

$$\xi_{ij}(0) = 1 + d_{ij} - \min(d_{ij}) \quad (8)$$

Where $d_{ij} = M_{ij} * \ell$ denotes the details within an image and ℓ is the 3 x 3 laplacian operator given by,

$$\ell = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Alsodenotes * convolution operator and M_{ij} is the normalized form of input image I whose intensity lies in the range of [0, 255]

The details in the image is determined by the mathematical equation defined in equation (9)

$$d_{ij} = \Lambda\left(\overline{M}_{ij} - M_{ij}\right)(1 - M_{ij}) \quad (9)$$

Where $\overline{M}_{ij}$ is a smoothed image of M_{ij} by the guided filtering [20], constant Λ is mainly controlling parameter of the magnitude details and is set 8 in our study.

After obtaining the details of the image from the mathematical equation (9) and applied it to the modified linking synaptic network, and obtained the synaptic output of the given image.

Further, the synaptic output is applied to the input of the modified contrast enhancement algorithm for the contrast enhancement of images. The modified contrast enhancement algorithm works in two parts i.e. one part operates for brighter regions and the other part works for darker region. For the contrast enhancement of brighter regions, a tune mapping function followed by edge-preserving illumination estimation is used and for the contrast enhancement in darker regions, the synaptic output with the perceptual contrast map is used [15]. After that there is adjustment of dynamic range of the conditioned output for the darker regions, so that edges get enhanced. Hence, enhanced output is obtained by the blending of above two outputs depending on a saliency map weighting and thereby preserving natural appearance [15].

b. Input Lightness

The synaptic output from the equation (1) after all neuron fired their spikes is applied for contrast enhancement. In order to enhance the contrast, the lightness component [15] is required and it is given by

$$R(x, y) = \max\{X(x, y)\} \quad (10)$$

Where $X(x, y)$ denotes the synaptic output which will be enhanced.

c. Detail Preserving Enhancement

The synaptic output is passed through this stage for the enhancement of contrast of image in bright regions. Here, the enhancement of bright regions obtained by the edge preserving tune mapping function proposed in [15]. This process does not introduce any unwanted circular artifacts. In order to compute edge preservation details, a gain parameter factor is defined at each pixel (i, j)

$$q_{ij} = 0.1\sqrt{\sigma(i, j)_{5\times5}} \quad (11)$$

Where $\sigma(i,j)_{5\times5}$ is the local standard deviation of a 5 × 5 neighbor window centered at (i, j) in the synaptic output image X.

Then, the image details are provided by scene reflectance which are given by

$$SR = \frac{R}{l+a} \tag{12}$$

Where l is the illumination component same as the mean luminance value of the synaptic and a is a small constant only to avoid zero division.

The details of the brighter regions get highlighted with the detail gain factor given as,

$$B_{en} = SR^{q} \tag{13}$$

This consists of all the dispensable halo artifacts and clipping effects. Hence, in order to remove them, we have applied a tune mapping function with B_{en} so that the contrast of bright regions can be improved [15] and it is given as

$$R_{bright_en} = R^{B_{en}} \tag{14}$$

The preservation of fine details during the improvement of contrast of bright regions and mid-tune range is guaranteed by tune mapping factor.

d. Auto perceptual contrast enhancement

The dark regions are left unchanged in details preserving contrast enhancement procedure because of low lightness [15]. A perceptual contrast map (PM) based on the modified difference of Gaussian (DOG) is used for extraction of edge contents [3][15]. The amplitude of the PM depicts the strength of information in the frequency. The construction of PM consists of central component $C_c(x,y)$ and surrounding component $S_c(x,y)$ are given by

$$C_c(x,y) = \sum_{i=x-r_c}^{i=x+r_c} \sum_{j=y-r_c}^{j=y+r_c} CG(i-x, j-y) I(i,j) \tag{15}$$

$$S_c(x,y) = \sum_{i=x-r_s}^{i=x+r_s} \sum_{j=y-r_s}^{j=y+r_s} SG(i-x, j-y) I(i,j) \tag{16}$$

Where $I(i,j)$ is the input image $r_s = 1$, $r_s = 0.01 * \min(h,w)$ and CG(x, y) and SG(x, y) are the two Gaussian functions given by

$$CG(x,y) = \exp\left[-\left(\frac{3x}{r_c}\right)^2 - \left(\frac{3y}{r_c}\right)^2\right] \tag{17}$$

$$SG(x,y) = 0.85\left(\frac{r_c}{r_s}\right)^2 \exp\left[-\left(\frac{3x}{r_c}\right)^2 - \left(\frac{3y}{r_c}\right)^2\right] \tag{18}$$

The conditioning of the synaptic output is done by

$$S(x,y) = R(x,y)\left|P(x,y)\right| \tag{19}$$

Where PM is defined by,

$$PM(x,y) = \frac{C_c(x,y) - S_c(x,y)}{S_c(x,y)} \tag{20}$$

e. Saliency guided blending

The salient regions contain fine details in well exposed regions and the saliency guided blending [15] equation for combination R_{bright_en} and $R_{percep_{en}}$ is given by

$$R_{en}(x,y) = Sal_{nor}(x,y) R_{bright_{en}} + (1 - Sal_{nor}(x,y)) R_{percep_{en}} \tag{21}$$

Where Sal_{nor} is the normalized saliency weight in the range of [0, 1]. It indicates that salient regions in R_{en} are enhanced with details preserving enhancement.

Finally, the enhanced component of lightness R_{en} is obtained and the resultant image of the proposed method is obtained with color correction in YUV space as

$$\left.\begin{aligned} r &= R_{en} / X \\ Y_{out} &= r(Y_{in} - 16) + 16 \\ U_{out} &= r.\alpha(U_{in} - 128) + 128 \\ V_{out} &= r.\alpha(V_{in} - 128) + 128 \end{aligned}\right\} \tag{22}$$

Where, X is the synaptic output, (Y_{in}, U_{in}, V_{in}) and $(Y_{out}, U_{out}, V_{out})$ depicts input and output images in YUV space respectively, and α is the color correction factor whose value oscillates between 1 to 2 with 1.8 as the default in our following experimentsDo not use the word "essentially" to mean "approximately" or "effectively."

III. Experimental Results And Discussion

The experimental results of proposed method is evaluated and compared with other existing enhancement methods like RHE-DCT [4], DWT-SVD [5], LCC [6], REINHARD [7], DRAGO [8] and LSCN [1]. The parameters of the above mentioned methods are set to the values as provided by the respective authors.

A. Quantitative and qualitative results assessment

For evaluating the performance of different remote sensing enhancement methods, we have considered five No-Reference qualitative metrics which are used in the remote sensing and other image processing domain. These includes NR Image Quality Metric forContrast distortion (NIQMC) [9] based on information maximization, Blind Image Quality Measure of Enhanced images (BIQME) [10] based on a measure of visual quality and learning with big data training samples, Colorfulness metric (CM) [11], Discrete Entropy (DE) [12], [13] and Pixel distance [14]. These metrics use default parameters as provided by the respective authors.

Test images (i.e. Image1 (1200x1200), Image2 (900x792)source:http://www.satpalda.com/gallery/pleiades-imagery/;Image3 (512x512), source:

http://visibleearth.nasa.gov/; Image4 (2000x1312), source: http://dragon.larc.nasa.gov/retinex/) are taken from the different sources for experimentation.

The quantitative results of proposed method and other existing enhancements algorithms for test Image1, Image2, Image3 and Image4 are given in Table 1, Table 2, Table 3 and Table 4, respectively. We have applied this proposed method to three remote sensing satellite images and one natural image from source mentioned above. In the experiments, we enhance the image and compare our proposed method with other enhancement methods. The qualitative results of proposed method and other existing algorithms for test images are given in Fig.1, Fig. 2, Fig. 3 and Fig.4 respectively.

TABLE I. PERFORMANCE COMPARISON OF DIFFERENT ENHANCEMENT TECHNIQUES FOR IMAGE 1

ALGORITHMS	METRICES				
	NIQMC	***BIQME***	***CM***	***DE***	***Pixdist***
DWT-SVD	3.6843	0.4195	44.1715	6.8093	15.6785
RHE-DCT	3.9025	0.5726	32.4546	6.8536	16.2517
LCC	4.6754	0.52	25.151	7.2827	24.3869
REINHARD	4.5928	0.5622	31.7501	7.0199	20.4145
DRAGO	4.4057	0.4262	29.877	6.9558	18.364
LSCN	5.1197	0.5432	25.3887	7.4286	25.4777
PROPOSED	**5.2574**	**0.5778**	**26.5101**	**7.5533**	**28.5136**

TABLE II. PERFORMANCE COMPARISON OF DIFFERENT ENHANCEMENT TECHNIQUES FOR IMAGE 2

ALGORITHMS	METRICES				
	NIQMC	***BIQME***	***CM***	***DE***	***Pixdist***
DWT-SVT	5.1595	0.5448	26.113	7.5371	28.1786
RHE-DCT	5.1342	0.6322	30.8111	7.5425	27.2012
LCC	5.0201	0.6006	31.485	7.5341	26.5717
REINHARD	4.9667	0.5039	36.4306	7.0752	27.288
DRAGO	4.4605	0.4322	36.9459	7.1843	23.0223
LSCN	5.2999	0.6151	32.1798	7.5532	28.921
PROPOSED	**5.4529**	**0.6558**	**26.7238**	**7.6227**	**33.8041**

TABLE III. PERFORMANCE COMPARISON OF DIFFERENT ENHANCEMENT TECHNIQUES FOR IMAGE 3

ALGORITHMS	METRICES				
	NIQMC	***BIQME***	***CM***	***DE***	***Pixdist***
DWT-SVT	4.7132	0.5112	39.6885	7.2083	24.8419
RHE-DCT	4.3532	0.5846	22.3648	6.823	17.8196
LCC	4.4001	0.5249	28.9765	7.0239	18.6426
REINHARD	4.0278	0.441	34.2199	6.726	23.127
DRAGO	4.8133	0.496	41.1393	7.383	26.8299
LSCN	4.741	0.5705	30.7833	7.2713	25.6841
PROPOSED	**5.1111**	**0.5684**	**56.5038**	**7.5914**	**29.754**

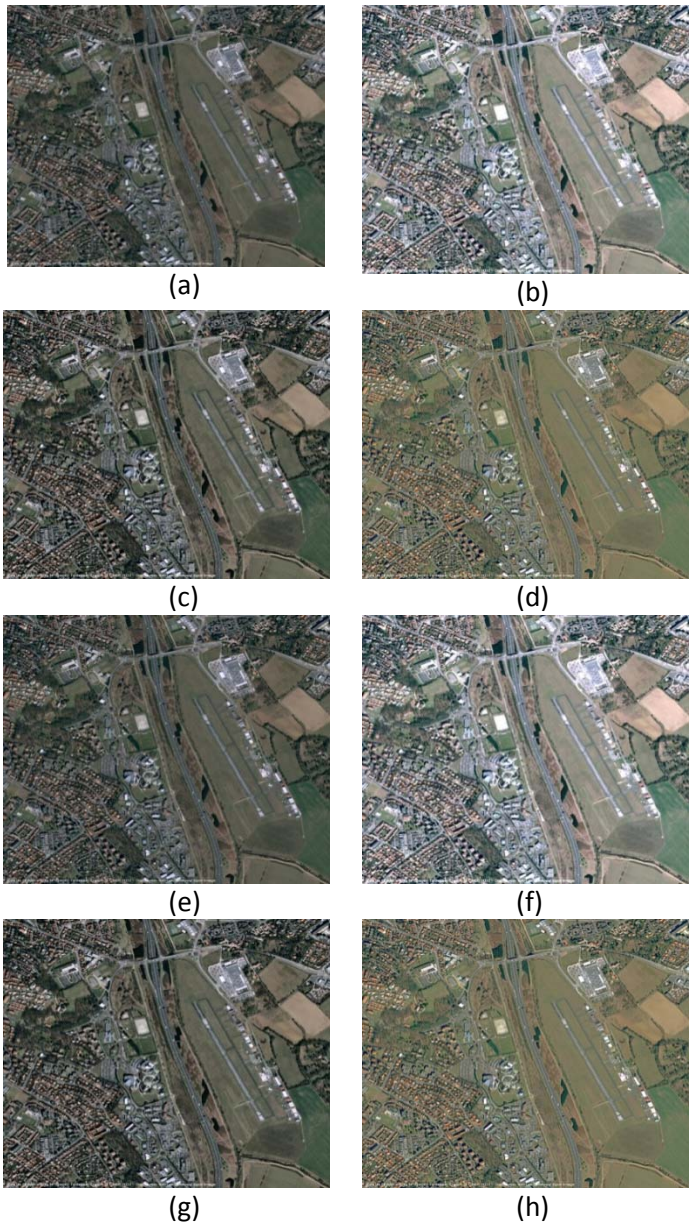

Fig. 1. Visual Enhancement Results of Different algorithms for image 1 (a) given, (b) DWT-SVT, (c) RHE-DCT, (d) LCC, (e) REINHARD, (f) DRAGO, (g) LSCN, (h) PROPOSED

TABLE IV. PERFORMANCE COMPARISON OF DIFFERENT ENHANCEMENT TECHNIQUES FOR IMAGE 4

ALGORITHMS	METRICES				
	NIQMC	***BIQME***	***CM***	***DE***	***Pixdist***
DWT-SVT	5.3379	0.6179	35.01396	7.5398	36.3637
RHE-DCT	5.068	0.5932	25.72515	7.4531	27.2026
LCC	3.7525	0.4652	56.7259	7.0239	18.6426
REINHARD	2.9719	0.4641	73.6201	6.5419	21.2633
DRAGO	2.9794	0.3185	43.7148	6.4215	12.5181
LSCN	4.8038	0.5866	43.4879	7.4499	24.5955
PROPOSED	**5.3863**	**0.6633**	**47.633**	**7.7211**	**31.0168**

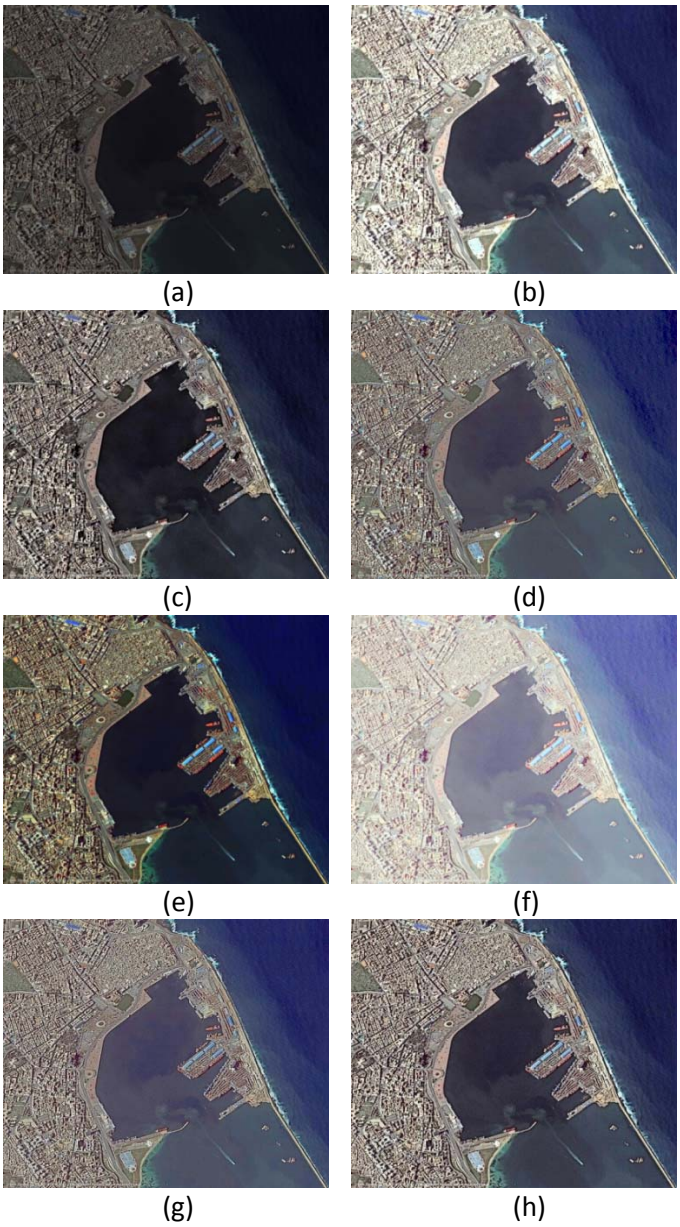

Fig. 2. Visual Enhancement Results of Different algorithms for image 2 (a) given, (b) DWT-SVT, (c) RHE-DCT, (d) LCC, (e) REINHARD, (f) DRAGO, (g) LSCN, (h) PROPOSED

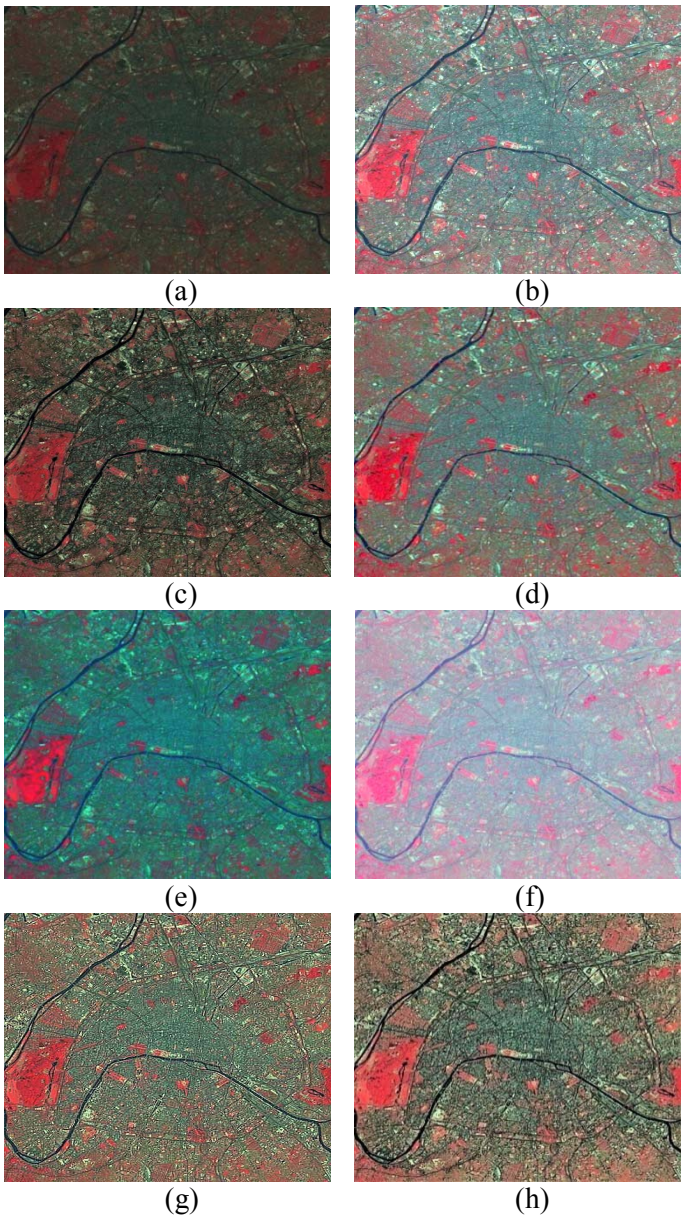

Fig. 3. Visual Enhancement Results of Different algorithms for image 3. (a) given, (b) DWT-SVT, (c) RHE-DCT, (d) LCC, (e) REINHARD, (f) DRAGO, (g) LSCN, (h) PROPOSED

B. Discussions

From the Figures 1(h), 2(h), 3(h), and 4(h) it is clear that for the given image, proposed method provides better visual enhancement results as compared with other existing remote sensing enhancement algorithms. Similarly, in RHE-SVD the enhanced details of the images cannot be distinguished clearly and also their contrast is low.

In LCC, there are losses of image details and in results the color appears slightly artificial. In REINHARD, the dark regions completely become dark leading to the loss of information. Also the naturalness is lost in this case. DRAGO causes excessive enhancement which results in overall whitening. In the case of LSCN, the results appear to be more smoothed and also less enhanced. But in case of proposed method, the results are a complete balanced in terms of contrast enhancement and preservation of naturalness. Therefore, we can say that proposed method provides better visual enhancement results as compared with other existing remote sensing enhancement algorithms. The CPU processing time of different enhancement algorithms is given in Table V.

TABLE V. CPU PROCESSING TIME (SECOND) OF DIFFERENT ALGORITHMS

Algorithm	Image1	Image2	Image3	Image4
RHE-DCT	1.4931	0.5812	0.4732	0.5565
LSCN	25.0595	7.008	5.0739	5.0499
PROPOSED	6.2252	2.5378	1.5216	5.9957

The computational complexity of proposed method less than LSCN algorithm for Image1, Image2, Image3 and

slightly more for Image 4, while processing time is little more than RHE-DCT method.

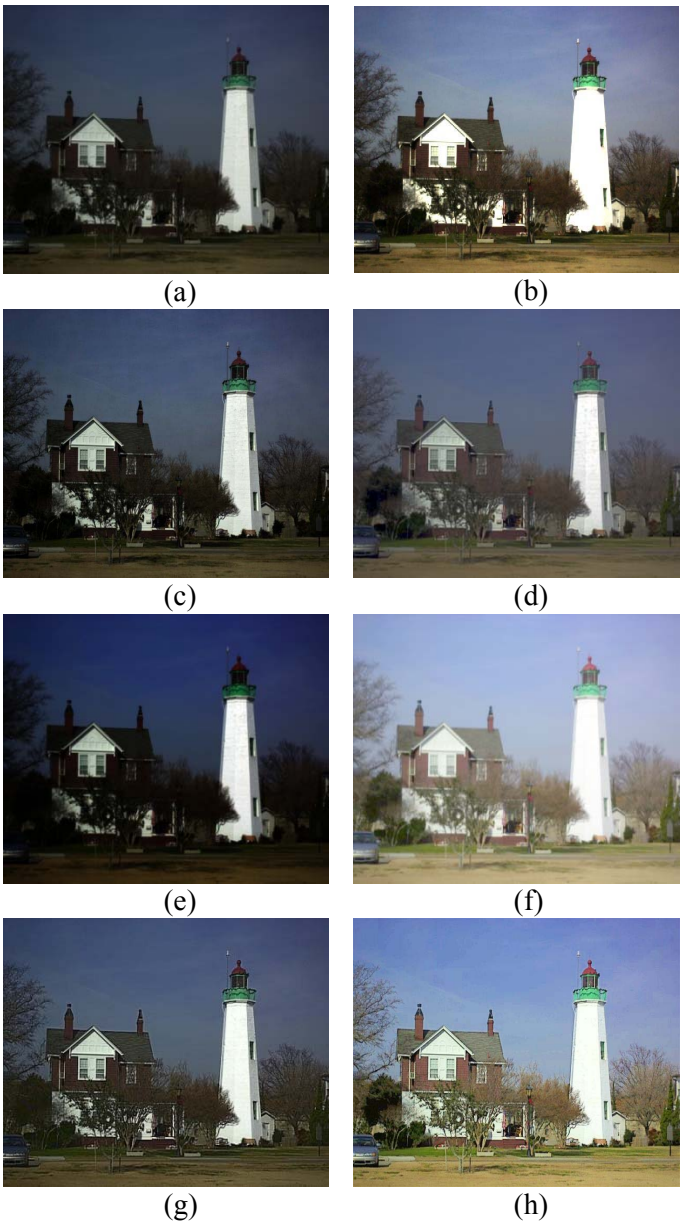

Fig. 4. Visual enhancement results of different algorithms for image 4. (a) given, (b) DWT-SVT, (c) RHE-DCT, (d) LCC, (e) REINHARD, (f) DRAGO, (g) LSCN, (h) PROPOSED

IV. Conclusion

This paper proposed a robust framework for contrast enhancement of multispectral remote sensing images. Performance metrics of different enhancement were computed in terms of DE, NIQMC, BIQME, PisDist and CM on different remote sensing and other image databases. Simulation results indicated that a quantitative result of proposed algorithm was better as compared to other existing remote sensing enhancement algorithms. A qualitative result also revealed that visual enhancement of the proposed method was better than other existing remote sensing enhancement algorithms. Finally, the simulation results shows that proposed algorithm is effective and efficient for remotes sensing as well as natural images.

References

[1]. Zhan, J. Shia, J. Teng, Q. Li, M. Wang, F. Lu, “Linking synaptic computation for image enhancement”, Neurocomputing, vol. 238, pp. 1-12, 2017.

[2]. A.S. French, R.B. Stein, “A flexible neural analog using integratedcircuits”, IEEE Trans. Biomed. Eng. BME, vol.17, no.3, pp. 248, 1970.

[3]. H. Zhang, Y. Li, H. Chen, D. Yuan, M. Sun, “Perceptual contrast enhancement with dynamic range adjustment”, Optik, vol. 124, no.23, pp. 5906-5913, 2013.

[4]. X. Fu, J. Wang, D. Zeng, Y. Huang, and X. Ding, “Remote Sensing Image Enhancement Using Regularized-Histogram Equalization and DCT”, IEEE Geosci. Remote Sens. Lett., vol. 12, no. 11, pp. 2301-2305, 2015.

[5]. H. Demirel, C. Ozcinar, and G. Anbarjafari, “Satellite image contrast enhancement using discrete wavelet transform and singular value decomposition”, IEEE Geosci. Remote Sens. Lett., vol. 7, no. 2, pp. 333–337, Apr. 2010

[6]. A. Moore, J. Allman, and R.M. Goodman, “A real-time neural system for color constancy”, IEEE Trans. Neural Netw., vol. 2, no. 2, pp. 237–246, Mar. 1991

[7]. E. Reinhard and K. Devlin, “Dynamic range reduction inspired by photoreceptor physiology”, IEEE Trans. Vis. Comput. Graph., vol. 11, no. 1, pp. 13–24, Feb. 2005.

[8]. F. Drago, K.Myszkowski, T. Annen, and N. Chiba, “Adaptive logarithmic mapping for displaying high contrast scenes”, Comput. Graph. Forum, vol. 22, no. 3, pp. 419–426, 2003.

[9]. K. Gu, W. Lin, G. Zhai, X. Yang, W. Zhang, and C. W. Chen, “No-Reference Quality Metric of Contrast-Distorted Images Based on Information Maximization” IEEE transaction on Cybernetics, 2017. DOI: 10.1109/TCYB.2016.2575544

[10]. K. Gu, D. Tao, Jun-Fei Qiao, W. Lin, “Learning a No-Reference Quality Assessment Model of Enhanced Images With Big Data”, IEEE Transactions On Neural Networks And Learning Systems, 2017. DOI: 10.1109/TNNLS.2017.2649101

[11]. D. Hasler and S. E. Suesstrunk, “Measuring colorfulness in natural images”, Proc. SPIE, vol. 5007, pp. 87–95, Jun. 2003.

[12]. C. E. Shannon, A mathematical theory of communication, Bell Syst. Tech. J., vol. 27, no. 3, pp. 379–423, Jul. 1948.

[13]. J. Shin and R. H. Park, “Histogram-based locality-preserving contrast enhancement”, IEEE Signal Process. Lett., vol. 22, no. 9, pp. 1293–1296, Sep. 2015.

[14]. Z. Chen, B. R. Abidi, D. L. Page, and M. A. Abidi, “Gray-level grouping (GLG): An automatic method for optimized image contrast enhancement-part I: The basic method”, IEEE Trans. Image Process., vol. 15, no. 8, pp. 2290–2302, Aug. 2006.

[15]. Y. Li, H. Zhanga, W. Jia, D. Yuana, F. Chenga, R. Jia, Lu Li, M. Sun, “Saliency guided naturalness enhancement in color images”, Optik, vol. 127, pp. 1326–1334, 2016.

[16]. R. Eckhorn, H. Reitboeck, M. Arndt, P. Dicke, “Feature linking via synchronization among distributed assemblies: simulations of results from cat visual cortex”, Neural Comput., vol. 2, no.3, p. 293, 1990.

[17]. J.L. Johnson, D. Ritter. “Observation of periodic waves in a pulse-coupled neural network”, Opt. Lett.,vol.18, no.15, p. 1253,1993.

[18]. J.L. Johnson, J.R. Taylor, M. Anderson, “Pulse-coupled neural network shadow compensation”, Proc. SPIE, 3722 (1999), p. 452.

[19]. J.L. Johnson, M.L. Padgett, “PCNN models and applications”, IEEE Trans. Neural Netw., vo. 10, no. 3, p. 480, 1990.

[20]. K.He, J. Sun, and X. Tang,”Guided Image Filtering”, IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 35, no.6, pp. 1397-1409, 2013.